

TIME-UNIFORM BOUNDEDNESS AND LARGE-TIME BEHAVIOR FOR A PLANAR DOUBLY DEGENERATE NUTRIENT-TAXIS SYSTEM

RUIYI ZHANG AND XIAOXIN ZHENG

ABSTRACT. We study the Cauchy problem for a doubly degenerate nutrient-taxis system in the plane. The initial population density is nonnegative and belongs to $L^1(\mathbb{R}^2) \cap L^\infty(\mathbb{R}^2)$. The initial nutrient concentration is nonnegative, nontrivial, integrable, and Lipschitz continuous, with finite weighted quartic gradient energy. We construct a global bounded weak solution whose population density and nutrient gradient are uniformly bounded in time. The estimates are derived from positive classical solutions of Neumann problems on expanding balls. A flux-aligned chemical entropy and the quartic logarithmic nutrient score, augmented by a multiple of the nutrient mass, form a compensated Lyapunov functional that controls the quartic score uniformly and the sixth logarithmic-gradient moment in spacetime. A radius-uniform endpoint inequality then induces the intrinsic exponent lift $p \mapsto 3p - 1$. Shifting its fixed point $1/2$ turns the cubic moment recursion into a convergent product and yields the population L^∞ -bound. A localized logarithmic entropy supplies positive-time H_{loc}^1 -regularity, and the flux estimates yield the compactness required by the Aubin–Lions lemma. We further prove positive-time local regularity, heat-semigroup decay of the nutrient in every $L^q(\mathbb{R}^2)$, $q > 1$, an $O(t^{-1})$ gradient bound, and convergence of the population in the bounded-Lipschitz dual norm and weakly in $L^1(\mathbb{R}^2)$.

1. INTRODUCTION

1.1. The model and related results. We consider

$$\begin{cases} u_t = \nabla \cdot (uv \nabla u) - \nabla \cdot (u^2 v \nabla v) + uv, & x \in \mathbb{R}^2, \quad t > 0, \\ v_t = \Delta v - uv, & x \in \mathbb{R}^2, \quad t > 0, \\ u(\cdot, 0) = u_0, \quad v(\cdot, 0) = v_0, & x \in \mathbb{R}^2. \end{cases} \quad (1.1)$$

Here u denotes the population density, and v denotes the concentration of a nutrient consumed by the population. Experiments on *Bacillus subtilis* colonies show pronounced changes in morphology under variations of the nutrient and agar concentrations [13, 11]. Leyva, Málaga, and Plaza proposed a nutrient-taxis model for these aggregation patterns [6]. Plaza later derived the system as a parabolic limit of a velocity-jump process [16].

System (1.1) is the quadratic case of a broader family in which the undirected and taxis fluxes have different density powers [21, 25]. The population diffusion degenerates at $u = 0$ and at $v = 0$, and nutrient consumption can drive v toward zero. Formally, sufficiently regular and decaying solutions satisfy $\frac{d}{dt} \int_{\mathbb{R}^2} (u + v) = 0$ and $\frac{d}{dt} \int_{\mathbb{R}^2} v = - \int_{\mathbb{R}^2} uv$. These relations suggest conservation of total mass and time-integrability of the consumption term. Both are proved below for the constructed solution. Their combination with weighted gradient estimates controls the two population fluxes.

On bounded one-dimensional intervals, weighted quasi-energies and weak-solution constructions for doubly degenerate nutrient systems were developed in [18, 8]. Two-dimensional bounded-domain results include small-signal solutions [19], general cross-diffusive fluxes [21], modified diffusion exponents [24], and the quadratic system [25]. Related extensions address higher-dimensional or large-data settings [7, 22], superlinear degradation or saturated consumption [14, 1], logistic proliferation [9, 10], and multispecies systems [23]. Three-dimensional bounded-domain results for subquadratic bacterial responses appear in [17, 4]. The taxis-free system admits stabilization of arbitrary structures in several dimensions [20].

For Cauchy problems, Herrero-Hervás constructed global weak solutions for the exact doubly degenerate system on the whole real line by expanding intervals and domain-uniform estimates [5].

2020 *Mathematics Subject Classification.* 35K65, 35K59, 35D30, 35B40, 92C17.

Key words and phrases. Doubly degenerate diffusion, nutrient taxis, compensated entropy, nutrient score, amplitude iteration, time-uniform boundedness, large-time behavior.

Cauchy problems for classical Keller–Segel systems have also been studied in critical spaces and in connection with planar global existence [3, 2, 12]. The present work treats the planar Cauchy problem for nonnegative integrable and bounded population data and allows the nutrient to vanish initially.

1.2. Main results and proof structure. The proof has five main steps. First, the chemical entropy generated by $\log(1 + ue^{-v})$ is combined with the quartic nutrient score $\int v|\nabla \log v|^4$. The reaction term in the score identity supplies a fourth-order weighted damping. Completing the common population-gradient terms produces the Schur square used in the closure, while the full logarithmic Hessian dissipation controls the sixth-order score with the sharp factor $28/3$. These two multipliers give

$$\sup_{t>0} \int_{\mathbb{R}^2} v|\nabla \log v|^4 + \int_0^\infty \int_{\mathbb{R}^2} v|\nabla \log v|^6 < \infty.$$

The weighted population-gradient estimate, the nutrient Dirichlet and Fisher estimates, and the remaining compactness bounds are then recovered from this two-multiplier closure.

Second, a radius-uniform endpoint inequality lifts the parent moment $\int_{B_R} f^r$ to $\int_{B_R} f^{3r}g$, paying a weighted population gradient and $\int_{B_R} |\nabla g|^6/g^5$. The exponent map $r \mapsto 3r - 1$ is forced by the Hölder factorization with exponents 6, 3, 2. It yields the affine cascade $p_{k+1} = 3p_k - 1$, starting from the mass exponent 1. Subtracting the fixed point $\frac{1}{2}$ conjugates this cascade to multiplication by 3 and turns the cubic moment recursion into a convergent product. Third, the resulting uniform bound gives global continuation on every expanding ball. A localized logarithmic entropy and local nutrient positivity then provide positive-time H_{loc}^1 -control of u .

Fourth, the local H^1 -bound and the global L^2 -flux estimates give the time compactness and nonlinear flux identification required for exhaustion. Finally, the heat representation gives the coefficient bounds used in the positive-time regularity argument. The full-time L^1 -integrability of both population fluxes and of uv yields the required time representative, traces, and mass balances. It also gives population tightness and the large-time conclusions. Sections 2 and 3 carry out the construction, while Sections 4 and 5 establish positive-time regularity and large-time behavior. Every radius-uniform tool and multiplier calculation is proved where it first enters the argument.

2. WEAK SOLUTIONS AND RADIUS-UNIFORM ESTIMATES

2.1. Notation and conventions. For $R \geq 1$, let $B_R = \{x \in \mathbb{R}^2 : |x| < R\}$. The domain of integration is stated explicitly in every spatial integral. In Sections 2 and 3, constants denoted by C may depend on $\|u_0\|_{L^1(\mathbb{R}^2)}$, $\|u_0\|_{L^\infty(\mathbb{R}^2)}$, $\|v_0\|_{L^1(\mathbb{R}^2)}$, $\|v_0\|_{L^\infty(\mathbb{R}^2)}$, and $\int_{\mathbb{R}^2} |\nabla v_0|^4/v_0^3$, and are independent of R and $T < T_{\max, R}$. The whole-space bound for ∇v also uses $\|\nabla v_0\|_{L^\infty(\mathbb{R}^2)}$. A subscript is added when a further parameter enters the constant.

For a nonnegative $v \in L^1(\mathbb{R}^2)$, we use the convention

$$\int_{\mathbb{R}^2} \frac{|\nabla v|^4}{v^3} := \begin{cases} 256 \int_{\mathbb{R}^2} |\nabla v^{1/4}|^4, & v^{1/4} \in W^{1,4}(\mathbb{R}^2), \\ +\infty, & v^{1/4} \notin W^{1,4}(\mathbb{R}^2). \end{cases} \quad (2.1)$$

When the first case applies, this agrees with the usual quotient on $\{v > 0\}$ and gives its lower semicontinuous extension at $v = 0$. Quotients involving v in the ball estimates have their usual meaning because the prepared ball solutions are positive.

We use the analogous lower semicontinuous convention

$$\int_{\mathbb{R}^2} \frac{|\nabla v|^6}{v^5} := \begin{cases} 6^6 \int_{\mathbb{R}^2} |\nabla v^{1/6}|^6, & v^{1/6} \in W^{1,6}(\mathbb{R}^2), \\ +\infty, & v^{1/6} \notin W^{1,6}(\mathbb{R}^2). \end{cases}$$

For a finite signed Radon measure μ on $\mathbb{R}^2$, define

$$\|\mu\|_{\text{BL}^*} = \sup_{\|\varphi\|_{W^{1,\infty}(\mathbb{R}^2)} \leq 1} \left| \int_{\mathbb{R}^2} \varphi d\mu \right|, \quad \|\varphi\|_{W^{1,\infty}} = \|\varphi\|_{L^\infty} + \|\nabla \varphi\|_{L^\infty}.$$

Functions in $L^1(\mathbb{R}^2)$ are identified with their absolutely continuous measures. We write $W_p^{2,1}(Q)$ for the parabolic Sobolev space with two spatial derivatives and one time derivative in $L^p(Q)$. All vector-valued

Lebesgue and Sobolev spaces are understood componentwise. For $0 < q < 1$, $\|\cdot\|_{L^q}$ denotes the usual L^q quasi-norm. We set $H^{-1}(\Omega) = (H_0^1(\Omega))^*$, and $|D^2\phi|$ denotes the Frobenius norm of a Hessian.

2.2. Solution concept and main results.

Definition 2.1. A pair (u, v) is a global bounded weak solution of (1.1) if $u \geq 0$, $v \geq 0$, and for every $T > 0$,

$$\begin{aligned} u &\in L^\infty(0, T; L^1(\mathbb{R}^2) \cap L^\infty(\mathbb{R}^2)), \\ v &\in L^\infty(0, T; L^1(\mathbb{R}^2) \cap W^{1,\infty}(\mathbb{R}^2)) \cap L^2(0, T; H^2(\mathbb{R}^2)), \\ v_t, uv &\in L^2(\mathbb{R}^2 \times (0, T)), \end{aligned} \quad (2.2)$$

and, for every bounded smooth domain $\Omega \Subset \mathbb{R}^2$ and $0 < \tau < T$,

$$u \in L^2(\tau, T; H^1(\Omega)), \quad u_t \in L^2(\tau, T; H^{-1}(\Omega)). \quad (2.3)$$

The two fluxes in the population equation satisfy $uv\nabla u$, $u^2v\nabla v \in L^2(\mathbb{R}^2 \times (0, T); \mathbb{R}^2)$. For every $\varphi, \psi \in C_c^\infty(\mathbb{R}^2 \times [0, T])$,

$$\int_0^T \int_{\mathbb{R}^2} u \varphi_t + \int_{\mathbb{R}^2} u_0 \varphi(\cdot, 0) = \int_0^T \int_{\mathbb{R}^2} (uv\nabla u - u^2v\nabla v) \cdot \nabla \varphi - \int_0^T \int_{\mathbb{R}^2} uv\varphi, \quad (2.4)$$

$$\int_0^T \int_{\mathbb{R}^2} v \psi_t + \int_{\mathbb{R}^2} v_0 \psi(\cdot, 0) = \int_0^T \int_{\mathbb{R}^2} \nabla v \cdot \nabla \psi + \int_0^T \int_{\mathbb{R}^2} uv\psi. \quad (2.5)$$

Theorem 2.2. Assume that

$$u_0 \geq 0, \quad u_0 \in L^1(\mathbb{R}^2) \cap L^\infty(\mathbb{R}^2),$$

and

$$v_0 \geq 0, \quad v_0 \not\equiv 0, \quad v_0 \in L^1(\mathbb{R}^2) \cap W^{1,\infty}(\mathbb{R}^2), \quad \int_{\mathbb{R}^2} \frac{|\nabla v_0|^4}{v_0^3} < \infty.$$

Then (1.1) possesses a global bounded weak solution (u, v) with representatives satisfying

$$u \in C([0, \infty); \text{BL}^*(\mathbb{R}^2)), \quad v \in C([0, \infty); L^1(\mathbb{R}^2)),$$

and

$$\text{ess sup}_{t>0} \left[\|u(\cdot, t)\|_{L^\infty(\mathbb{R}^2)} + \|\nabla v(\cdot, t)\|_{L^\infty(\mathbb{R}^2)} + \int_{\mathbb{R}^2} (u+v)(x, t) dx \right] \leq C.$$

Here C depends only on

$$\|u_0\|_{L^1}, \quad \|u_0\|_{L^\infty}, \quad \|v_0\|_{L^1}, \quad \|v_0\|_{L^\infty}, \quad \|\nabla v_0\|_{L^\infty}, \quad \int_{\mathbb{R}^2} \frac{|\nabla v_0|^4}{v_0^3}.$$

The solution also satisfies

$$0 \leq v \leq \|v_0\|_{L^\infty(\mathbb{R}^2)} \quad \text{a.e. in } \mathbb{R}^2 \times (0, \infty),$$

and for every compact $K \Subset \mathbb{R}^2$ and $0 < \tau < T < \infty$,

$$\text{ess inf}_{K \times (\tau, T)} v > 0. \quad (2.6)$$

The initial data are attained in the following senses:

$$\|u(\cdot, t) - u_0\|_{\text{BL}^*} \longrightarrow 0, \quad \|v(\cdot, t) - v_0\|_{L^1(\mathbb{R}^2)} \longrightarrow 0 \quad \text{as } t \downarrow 0. \quad (2.7)$$

For every $t \geq 0$,

$$\int_{\mathbb{R}^2} v(x, t) dx + \int_0^t \int_{\mathbb{R}^2} uv = \int_{\mathbb{R}^2} v_0, \quad (2.8)$$

$$\int_{\mathbb{R}^2} u(x, t) dx = \int_{\mathbb{R}^2} u_0 + \int_0^t \int_{\mathbb{R}^2} uv, \quad (2.9)$$

$$\int_{\mathbb{R}^2} (u+v)(x, t) dx = \int_{\mathbb{R}^2} (u_0 + v_0). \quad (2.10)$$

Proposition 2.3. *The solution constructed in Theorem 2.2 satisfies*

$$\operatorname{ess\,sup}_{t>0} \int_{\mathbb{R}^2} \frac{|\nabla v|^4}{v^3} + \int_0^\infty \int_{\mathbb{R}^2} \left(uv + u^2 v + \frac{|\nabla v|^6}{v^5} + |\Delta v|^2 \right) \leq C, \quad (2.11)$$

$$\int_0^\infty \int_{\mathbb{R}^2} \left(uv |\nabla u|^2 + u^3 v |\nabla v|^2 + |uv \nabla u|^2 + |u^2 v \nabla v|^2 + (uv)^2 \right) \leq C. \quad (2.12)$$

The two population fluxes also belong to $L^1(\mathbb{R}^2 \times (0, \infty))$.

Theorem 2.4 (Positive-time regularity). *For every bounded smooth domain $\Omega \Subset \mathbb{R}^2$ and every $0 < \tau < T < \infty$,*

$$u \in L^2(\tau, T; H^1(\Omega)), \quad u_t \in L^2(\tau, T; H^{-1}(\Omega)), \quad v \in W_p^{2,1}(\Omega \times (\tau, T)) \quad \text{for every } 1 < p < \infty.$$

There exist $\alpha \in (0, 1)$ and $C_{\Omega, \tau, T} > 0$, where α may depend on Ω, τ, T and on the structural bounds determined by the initial data, such that

$$\|u\|_{C^{\alpha, \alpha/2}(\bar{\Omega} \times [\tau, T])} + \|v\|_{C^{2+\alpha, 1+\alpha/2}(\bar{\Omega} \times [\tau, T])} \leq C_{\Omega, \tau, T}.$$

Theorem 2.5 (Large-time behavior). *For every $1 < q \leq \infty$,*

$$\|v(\cdot, t)\|_{L^q(\mathbb{R}^2)} \leq C_q t^{-(1-1/q)} \|v_0\|_{L^1(\mathbb{R}^2)}, \quad t > 0, \quad (2.13)$$

and

$$\|\nabla v(\cdot, t)\|_{L^\infty(\mathbb{R}^2)} \leq \frac{C}{1+t}, \quad t > 0. \quad (2.14)$$

There is a nonnegative $u_\infty \in L^1(\mathbb{R}^2) \cap L^\infty(\mathbb{R}^2)$ such that

$$u(\cdot, t) \rightharpoonup u_\infty \quad \text{weakly in } L^1(\mathbb{R}^2) \quad \text{as } t \rightarrow \infty, \quad (2.15)$$

and

$$\|u(\cdot, t) - u_\infty\|_{\text{BL}^*} \rightarrow 0. \quad (2.16)$$

The limiting masses satisfy

$$\int_{\mathbb{R}^2} u_\infty = \int_{\mathbb{R}^2} u_0 + \int_0^\infty \int_{\mathbb{R}^2} uv, \quad (2.17)$$

$$\lim_{t \rightarrow \infty} \int_{\mathbb{R}^2} v(\cdot, t) = \int_{\mathbb{R}^2} v_0 - \int_0^\infty \int_{\mathbb{R}^2} uv \geq 0. \quad (2.18)$$

2.3. Preparation of the initial data. Choose $R_j \rightarrow \infty$ with $R_j > 4$. Let $\chi_j \in C_c^\infty(B_{R_j-2}; [0, 1])$ satisfy $\chi_j = 1$ on B_{R_j-3} and $|\nabla \chi_j| \leq C$. Let $\rho \in C_c^\infty(B_1)$ be a nonnegative mollifier with $\int_{\mathbb{R}^2} \rho = 1$, and set $\rho_h(x) = h^{-2} \rho(x/h)$.

Lemma 2.6. *There exist positive functions $u_{0,j}, v_{0,j} \in C^\infty(\overline{B_{R_j}})$ satisfying $\partial_\nu u_{0,j} = \partial_\nu v_{0,j} = 0$ on ∂B_{R_j} , such that*

$$u_{0,j} \rightarrow u_0, \quad v_{0,j} \rightarrow v_0 \quad \text{in } L^1(\mathbb{R}^2)$$

after zero extension outside B_{R_j} , and

$$\begin{aligned} \sup_j \int_{B_{R_j}} (u_{0,j} + v_{0,j}) &< \infty, \\ \sup_j \|u_{0,j}\|_{L^\infty(B_{R_j})} &< \infty, \quad 0 < v_{0,j} \leq \|v_0\|_{L^\infty(\mathbb{R}^2)}, \\ \sup_j \int_{B_{R_j}} \frac{|\nabla v_{0,j}|^4}{v_{0,j}^3} &< \infty. \end{aligned} \quad (2.19)$$

There exists a single constant $C_0 \geq 1$, independent of p and j , such that, for every $p \geq 1$,

$$\sup_j \int_{B_{R_j}} u_{0,j}^p \leq C_0^p. \quad (2.20)$$

Proof. Choose $\varepsilon_j \downarrow 0$ and $h_j \downarrow 0$ so that $\varepsilon_j |B_{R_j}| \rightarrow 0$, and define

$$u_{0,j} = \varepsilon_j + \rho_{h_j} * (\chi_j u_0) \quad \text{in } B_{R_j}, \quad h_j < 1.$$

The convolution term vanishes near ∂B_{R_j} , so $u_{0,j}$ is positive and constant there. Convolution, the $L^1 \cap L^\infty$ -bounds for u_0 , and $\varepsilon_j \leq 1$ give the mass and L^∞ -bounds and a single $C_0 \geq 1$ such that (2.20) holds for every $p \geq 1$. Decreasing h_j , if needed, we may also require

$$\|\rho_{h_j} * (\chi_j u_0) - \chi_j u_0\|_{L^1(\mathbb{R}^2)} \leq j^{-1}.$$

Since $\chi_j u_0 \rightarrow u_0$ in L^1 , this proves the asserted convergence after zero extension.

By (2.1), $v_0^{1/4} \in W^{1,4}(\mathbb{R}^2) \cap L^4(\mathbb{R}^2)$. Choose $0 < \sigma_j < \|v_0\|_{L^\infty}^{1/4}$ with $\sigma_j^4 |B_{R_j}| \rightarrow 0$, and put

$$v_{0,j} = \left[\sigma_j + \left(1 - \frac{\sigma_j}{\|v_0\|_{L^\infty}^{1/4}} \right) \rho_{h_j} * (\chi_j v_0^{1/4}) \right]^4.$$

Then $0 < v_{0,j}^{1/4} \leq \|v_0\|_{L^\infty}^{1/4}$, and $v_{0,j}^{1/4}$ is constant near ∂B_{R_j} . Thus $v_{0,j}$ is positive, satisfies the Neumann condition, and is bounded by $\|v_0\|_{L^\infty}$. A further decrease of h_j gives

$$v_{0,j}^{1/4} \rightarrow v_0^{1/4} \quad \text{in } L^4(\mathbb{R}^2)$$

after zero extension. Indeed, the cutoff and mollification errors vanish in L^4 , while the constant part is controlled by $\sigma_j^4 |B_{R_j}|$. Consequently,

$$\|v_{0,j} - v_0\|_{L^1} \leq 4(\|v_{0,j}^{1/4}\|_{L^4} + \|v_0^{1/4}\|_{L^4})^3 \|v_{0,j}^{1/4} - v_0^{1/4}\|_{L^4} \rightarrow 0.$$

Finally, Young's convolution inequality and $\nabla(\chi_j v_0^{1/4}) = \chi_j \nabla v_0^{1/4} + v_0^{1/4} \nabla \chi_j$ give

$$\sup_j \int_{B_{R_j}} v_{0,j} < \infty, \quad \sup_j \int_{B_{R_j}} |\nabla v_{0,j}^{1/4}|^4 \leq C \int_{\mathbb{R}^2} (|\nabla v_0^{1/4}|^4 + v_0).$$

The identity

$$\frac{|\nabla v_{0,j}|^4}{v_{0,j}^3} = 256 |\nabla v_{0,j}^{1/4}|^4$$

proves (2.19). □

Hölder's inequality, the mass bound, and (2.19) also give

$$\sup_j \left(\int_{B_{R_j}} \frac{|\nabla v_{0,j}|^2}{v_{0,j}} + \int_{B_{R_j}} |\nabla v_{0,j}|^2 \right) < \infty. \quad (2.21)$$

2.4. Positive solutions on expanding balls and two radius-uniform inequalities. For a prepared pair on B_R , consider

$$\begin{cases} u_t = \nabla \cdot (uv \nabla u) - \nabla \cdot (u^2 v \nabla v) + uv, \\ v_t = \Delta v - uv, \\ \partial_\nu u = \partial_\nu v = 0 \quad \text{on } \partial B_R, \\ u(\cdot, 0) = u_{0,R}, \quad v(\cdot, 0) = v_{0,R}. \end{cases} \quad (2.22)$$

Lemma 2.7. *Problem (2.22) has a positive classical solution on a maximal interval $[0, T_{\max,R})$, and*

$$T_{\max,R} < \infty \implies \limsup_{t \uparrow T_{\max,R}} \|u(\cdot, t)\|_{L^\infty(B_R)} = \infty. \quad (2.23)$$

For every $0 < T < T_{\max,R}$,

$$\begin{aligned} \int_{B_R} (u + v)(t) &= \int_{B_R} (u_{0,R} + v_{0,R}), \\ \frac{d}{dt} \int_{B_R} v &= - \int_{B_R} uv, \quad \int_0^T \int_{B_R} uv \leq \int_{B_R} v_{0,R}, \\ 0 < v(x, t) &\leq \|v_0\|_{L^\infty(\mathbb{R}^2)}. \end{aligned} \quad (2.24)$$

Proof. By [25, Lemma 2.1], the fixed-domain problem has a local positive classical solution and satisfies the stated extensibility criterion. The prepared data meet the smoothness, positivity, and Neumann compatibility assumptions. Testing the two equations by one gives the mass identities and consumption bound; the maximum principle gives the pointwise estimate for v . $\square$

Lemma 2.8. *Let $0 < q \leq 1$. There is $C_q > 0$ such that every nonnegative $f \in W^{1,1}(B_R)$ satisfies*

$$\|f\|_{L^2(B_R)} \leq C_q \|\nabla f\|_{L^1(B_R)} + C_q R^{1-2/q} \|f\|_{L^q(B_R)}. \quad (2.25)$$

The constants C_q may be chosen uniformly when q ranges in a compact subinterval of $(0, 1]$.

Proof. On B_1 , let

$$\ell = \left(\frac{2}{|B_1|} \int_{B_1} f^q \right)^{1/q}.$$

Then $|\{f > \ell\}| \leq |B_1|/2$. The endpoint Sobolev–Poincaré inequality applied to $(f - \ell)_+$, which vanishes on a set of measure at least $|B_1|/2$, gives

$$\|(f - \ell)_+\|_{L^2(B_1)} \leq C \|\nabla f\|_{L^1(B_1)}.$$

Since $f \leq (f - \ell)_+ + \ell$,

$$\|f\|_{L^2(B_1)} \leq C \|\nabla f\|_{L^1(B_1)} + C_q \|f\|_{L^q(B_1)}.$$

The displayed choice of ℓ shows that C_q is bounded when q ranges in a compact subinterval of $(0, 1]$. Applying this inequality to $f(R \cdot)$ proves (2.25), including the factor $R^{1-2/q}$. $\square$

Lemma 2.9. *Let $v > 0$ be smooth in B_R with $\partial_\nu v = 0$. For every $q \geq 2$,*

$$\int_{B_R} v |\nabla \log v|^{q+2} \leq (q + \sqrt{2})^2 \int_{B_R} v |\nabla \log v|^{q-2} |D^2 \log v|^2. \quad (2.26)$$

Consequently,

$$\int_{B_R} \frac{|\Delta v|^2}{v} \leq (16 + 8\sqrt{2}) \int_{B_R} v |D^2 \log v|^2, \quad (2.27)$$

$$\int_{B_R} \frac{|\nabla v|^6}{v^5} \leq \frac{28}{3} \left[\int_{B_R} v |\nabla \log v|^2 |D^2 \log v|^2 + 2 \int_{B_R} v |D^2 \log v| |\nabla \log v|^2 \right]. \quad (2.28)$$

Proof. Since $\nabla \log v \cdot \nu = 0$ on ∂B_R , integration by parts and $|\Delta \log v| \leq \sqrt{2} |D^2 \log v|$ give

$$\begin{aligned} \int_{B_R} v |\nabla \log v|^{q+2} &= - \int_{B_R} v |\nabla \log v|^q \Delta \log v - q \int_{B_R} v |\nabla \log v|^{q-2} D^2 \log v [\nabla \log v, \nabla \log v] \\ &\leq (q + \sqrt{2}) \left(\int_{B_R} v |\nabla \log v|^{q+2} \right)^{1/2} \left(\int_{B_R} v |\nabla \log v|^{q-2} |D^2 \log v|^2 \right)^{1/2}. \end{aligned}$$

This proves (2.26). Moreover,

$$\frac{\Delta v}{v} = \Delta \log v + |\nabla \log v|^2.$$

Using $|\Delta \log v|^2 \leq 2 |D^2 \log v|^2$ and (2.26) with $q = 2$, we obtain

$$\begin{aligned} \int_{B_R} \frac{|\Delta v|^2}{v} &\leq 4 \int_{B_R} v |D^2 \log v|^2 + 2(2 + \sqrt{2})^2 \int_{B_R} v |D^2 \log v|^2 \\ &= (16 + 8\sqrt{2}) \int_{B_R} v |D^2 \log v|^2, \end{aligned}$$

which is (2.27). For the sharper quartic consequence, a second integration by parts gives

$$\int_{B_R} v |\nabla \log v|^6 = - \int_{B_R} v |\nabla \log v|^4 \left(\Delta \log v + 4 D^2 \log v \left[\frac{\nabla \log v}{|\nabla \log v|}, \frac{\nabla \log v}{|\nabla \log v|} \right] \right).$$

The bracketed directional term is set to zero where $\nabla \log v = 0$. At every point with $\nabla \log v \neq 0$, choose an orthonormal frame whose first vector is parallel to $\nabla \log v$. Then

$$(5\partial_{11} \log v + \partial_{22} \log v)^2 \leq \frac{28}{3} [3|\partial_{11} \log v|^2 + 4|\partial_{12} \log v|^2 + |\partial_{22} \log v|^2],$$

and the constant $28/3$ is the squared dual norm of $(5, 0, 1)$ for this quadratic form. Cauchy's inequality therefore yields (2.28); the set where $\nabla \log v = 0$ follows by continuity. $\square$

2.5. Entropy and weighted nutrient estimates. We use the elementary bounds

$$\log^2(1+s) \leq s, \quad \log^3(1+s) \leq 2s, \quad \left(2\log(1+s) - \frac{s}{1+s}\right)^2 \leq 4s, \quad s \geq 0, \quad (2.29)$$

and the exact identity

$$s - 2\log(1+s) + \frac{s}{1+s} = \int_0^s \frac{\sigma^2}{(1+\sigma)^2} d\sigma \geq 0.$$

The multiplier pair can be read directly from the two fluxes. The population flux has the aligned direction $\nabla u - u\nabla v = e^v \nabla(ue^{-v})$, so every scalar multiplier that turns this flux into a nonnegative quadratic form has the form $\phi(ue^{-v})$. The quartic score calculation also uses the exact splitting

$$\nabla u = \nabla \phi(u) + (1 - \phi'(u))(\nabla u - u\nabla v) + u(1 - \phi'(u))\nabla v.$$

Requiring the two aligned coefficients to have the same profile gives $s\phi'(s) = 1 - \phi'(s)$. With $\phi(0) = 0$, this compatibility selects $\phi(s) = \log(1+s)$ within the aligned family. Its primitive is

$$\int_0^u \log(1 + se^{-v}) ds = e^v [(1 + ue^{-v}) \log(1 + ue^{-v}) - ue^{-v}],$$

and the corresponding splitting becomes

$$\nabla u = \nabla \log(1+u) + \frac{u}{1+u}(\nabla u - u\nabla v) + \frac{u^2}{1+u}\nabla v.$$

The fourth-order nutrient score is the unique meeting point of the present one-score closure. Indeed, replacing $|\nabla \log v|^4$ by $|\nabla \log v|^m$ makes the Schur remainder grow like $|\nabla \log v|^{2m-2}$, while the logarithmic Bochner identity supplies only the $(m+2)$ -order score, so this completion requires $m \leq 4$. The population amplification below requires the instantaneous fourth-order score and the time-integrated sixth-order score, which gives $m \geq 4$. Thus $m = 4$ is forced within this Schur–Bochner–endpoint mechanism.

Lemma 2.10 (Adjustable population–nutrient bridge). *For every $\varepsilon, \eta > 0$,*

$$\int_{B_R} u^2 v \leq \varepsilon \int_{B_R} v \frac{u}{e^v + u} |\nabla u - u\nabla v|^2 + \eta \int_{B_R} uv |\nabla \log v|^4 + C_{\varepsilon, \eta} \int_{B_R} uv. \quad (2.30)$$

Proof. For $\ell \geq 1$, the sharp pointwise splitting

$$u^2 \leq \ell u + \frac{32}{27} \left(\frac{u^2}{\ell + u} \right)^2$$

has equality at $u = 3\ell$. Hence

$$\int_{B_R} u^2 v \leq \ell \int_{B_R} uv + \frac{32}{27} \int_{B_R} \frac{u^4 v}{(\ell + u)^2}. \quad (2.31)$$

Direct differentiation gives

$$\begin{aligned} \nabla \left(\frac{u^2 \sqrt{v}}{\ell + u} \right) &= \frac{u(2\ell + u)}{(\ell + u)^2} \sqrt{v} (\nabla u - u\nabla v) \\ &\quad + \left[\frac{u^2(2\ell + u)}{(\ell + u)^2} \sqrt{v} + \frac{u^2}{2(\ell + u)\sqrt{v}} \right] \nabla v. \end{aligned}$$

The coefficient bounds

$$\frac{u(2\ell + u)^2(e^v + u)}{(\ell + u)^4} \leq 4e^{\|v_0\|_{L^\infty}} \frac{u}{\ell}, \quad \frac{1}{u} \left[\frac{u^2(2\ell + u)v}{(\ell + u)^2} + \frac{u^2}{2(\ell + u)} \right]^2 \leq Cu$$

and Cauchy–Schwarz yield

$$\begin{aligned} \left(\int_{B_R} \frac{u(2\ell + u)}{(\ell + u)^2} \sqrt{v} |\nabla u - u\nabla v| \right)^2 &\leq \frac{C}{\ell} \int_{B_R} v \frac{u}{e^v + u} |\nabla u - u\nabla v|^2, \\ \left(\int_{B_R} \left[\frac{u^2(2\ell + u)}{(\ell + u)^2} \sqrt{v} + \frac{u^2}{2(\ell + u)\sqrt{v}} \right] |\nabla v| \right)^2 &\leq C \int_{B_R} \frac{u}{v} |\nabla v|^2. \end{aligned}$$

Moreover,

$$\left(\int_{B_R} \frac{u^2 \sqrt{v}}{\ell + u} \right)^2 \leq \left(\int_{B_R} u \right) \left(\int_{B_R} uv \right) \leq C \int_{B_R} uv.$$

Lemma 2.8 with $q = 1$, followed by (2.31), gives

$$\int_{B_R} u^2 v \leq \frac{C}{\ell} \int_{B_R} v \frac{u}{e^v + u} |\nabla u - u \nabla v|^2 + C \int_{B_R} \frac{u}{v} |\nabla v|^2 + C(\ell + 1) \int_{B_R} uv. \quad (2.32)$$

For every $\sigma > 0$, exact scalar optimization gives

$$\int_{B_R} \frac{u}{v} |\nabla v|^2 \leq \sigma \int_{B_R} uv |\nabla \log v|^4 + \frac{1}{4\sigma} \int_{B_R} uv. \quad (2.33)$$

Choose ℓ so that $C/\ell \leq \varepsilon$, and then choose σ so that $C\sigma \leq \eta$. This proves (2.30). $\square$

For a ball solution, denote the two multiplier energies by

$$\mathfrak{e}(t) = \int_{B_R} e^v [(1 + ue^{-v}) \log(1 + ue^{-v}) - ue^{-v}], \quad (2.34)$$

$$\mathfrak{q}(t) = \int_{B_R} \frac{|\nabla v|^4}{v^3}. \quad (2.35)$$

Lemma 2.11 (Two-multiplier Schur closure). *There are $\delta > 0$ and $C_\delta > 0$, independent of R and time, such that*

$$\begin{aligned} \frac{d}{dt}(\mathfrak{e} + \delta \mathfrak{q}) + \frac{1}{2} \int_{B_R} v \frac{u}{e^v + u} \left| \nabla u - u \nabla v - \left[\frac{1}{2} - 2\delta \frac{e^v + u}{1 + u} |\nabla \log v|^2 \right] \nabla \log v \right|^2 \\ + \delta \left[\int_{B_R} v |\nabla \log v|^2 |D^2 \log v|^2 + 2 \int_{B_R} v |D^2 \log v \nabla \log v|^2 \right] \\ + \frac{9\delta}{112} \int_{B_R} \frac{|\nabla v|^6}{v^5} + \frac{\delta}{2} \int_{B_R} uv |\nabla \log v|^4 \leq C_\delta \int_{B_R} uv. \end{aligned} \quad (2.36)$$

Moreover,

$$\begin{aligned} \int_{B_R} v \frac{u}{e^v + u} |\nabla u - u \nabla v|^2 \leq 2 \int_{B_R} v \frac{u}{e^v + u} \left| \nabla u - u \nabla v - \left[\frac{1}{2} - 2\delta \frac{e^v + u}{1 + u} |\nabla \log v|^2 \right] \nabla \log v \right|^2 \\ + \frac{1}{64\delta} \int_{B_R} uv + 8\Lambda_0 \delta^2 \int_{B_R} \frac{|\nabla v|^6}{v^5}, \end{aligned} \quad (2.37)$$

where

$$\Lambda_0 = \sup_{s \geq 0} \frac{s(e^{\|v_0\|_{L^\infty}} + s)}{(1 + s)^2} = \begin{cases} 1, & e^{\|v_0\|_{L^\infty}} \leq 2, \\ \frac{e^{2\|v_0\|_{L^\infty}}}{4(e^{\|v_0\|_{L^\infty}} - 1)}, & e^{\|v_0\|_{L^\infty}} > 2. \end{cases} \quad (2.38)$$

Proof. Differentiating the density in (2.34) with respect to u and v gives

$$\log(1 + ue^{-v}), \quad e^v [\log(1 + ue^{-v}) - ue^{-v}],$$

respectively. Substitution of the two equations and integration by parts give

$$\begin{aligned} \mathfrak{e}' + \int_{B_R} v \frac{u}{e^v + u} |\nabla u - u \nabla v|^2 = \int_{B_R} \frac{u}{e^v + u} (\nabla u - u \nabla v) \cdot \nabla v + \int_{B_R} e^v [ue^{-v} - \log(1 + ue^{-v})] |\nabla v|^2 \\ + \int_{B_R} uv [u - (e^v - 1) \log(1 + ue^{-v})]. \end{aligned} \quad (2.39)$$

The pointwise bounds

$$0 \leq e^v [ue^{-v} - \log(1 + ue^{-v})] \leq u, \quad 0 \leq u - (e^v - 1) \log(1 + ue^{-v}) \leq u$$

give

$$\int_{B_R} e^v [ue^{-v} - \log(1 + ue^{-v})] |\nabla v|^2 \leq \|v_0\|_{L^\infty} \int_{B_R} \frac{u}{v} |\nabla v|^2,$$

while the final term in (2.39) is bounded by $\int_{B_R} u^2 v$.

The identity

$$(\log v)_t = \Delta \log v + |\nabla \log v|^2 - u$$

and the quartic Bochner formula give

$$\begin{aligned} \mathfrak{q}' + 4 \int_{B_R} v |\nabla \log v|^2 |D^2 \log v|^2 + 8 \int_{B_R} v |D^2 \log v \nabla \log v|^2 + \frac{4}{R} \int_{\partial B_R} v |\nabla \log v|^4 + \int_{B_R} uv |\nabla \log v|^4 \\ = -4 \int_{B_R} v |\nabla \log v|^2 \nabla \log v \cdot \nabla u. \end{aligned} \quad (2.40)$$

The Neumann condition and the curvature $1/R$ of the circle produce the nonnegative boundary term. Inserting the exact decomposition of ∇u displayed above, the last summand gives a nonpositive contribution, and the term containing $\nabla u - u\nabla v$ is retained for the joint completion.

For the logarithmic term, integration by parts uses

$$\nabla \cdot (v |\nabla \log v|^2 \nabla \log v) = v |\nabla \log v|^4 + v |\nabla \log v|^2 \Delta \log v + 2v D^2 \log v [\nabla \log v, \nabla \log v].$$

Its boundary term vanishes because $\nabla \log v \cdot \nu = 0$. At a point where $\nabla \log v \neq 0$, choose an orthonormal frame whose first vector is parallel to $\nabla \log v$. The logarithmic term is bounded above by

$$4 \int_{B_R} v \log(1+u) |\nabla \log v|^4 + 4 \int_{B_R} v \log(1+u) |\nabla \log v|^2 |3\partial_{11} \log v + \partial_{22} \log v|.$$

For $\beta, \alpha > 0$, exact scalar optimization gives

$$4 \log(1+u) |\nabla \log v|^4 \leq \beta |\nabla \log v|^6 + \frac{256}{27\beta^2} \log^3(1+u),$$

$$4 \log(1+u) |3\partial_{11} \log v + \partial_{22} \log v| \leq \alpha \left[3|\partial_{11} \log v|^2 + 4|\partial_{12} \log v|^2 + |\partial_{22} \log v|^2 \right] + \frac{16}{\alpha} \log^2(1+u).$$

Together with (2.29), these estimates yield

$$\begin{aligned} 4 \left| \int_{B_R} v |\nabla \log v|^2 \nabla \log v \cdot \nabla \log(1+u) \right| \leq \beta \int_{B_R} v |\nabla \log v|^6 \\ + \alpha \left[\int_{B_R} v |\nabla \log v|^2 |D^2 \log v|^2 + 2 \int_{B_R} v |D^2 \log v \nabla \log v|^2 \right] \\ + \frac{16}{\alpha} \int_{B_R} \frac{u}{v} |\nabla v|^2 + \frac{512}{27\beta^2} \int_{B_R} uv. \end{aligned} \quad (2.41)$$

Add (2.39) to δ times (2.40). The two terms containing $\nabla u - u\nabla v$ have the exact completion

$$\begin{aligned} v \frac{u}{e^v + u} |\nabla u - u\nabla v|^2 - v \left[\frac{u}{e^v + u} - 4\delta \frac{u}{1+u} |\nabla \log v|^2 \right] (\nabla u - u\nabla v) \cdot \nabla \log v \\ = v \frac{u}{e^v + u} \left| \nabla u - u\nabla v - \left[\frac{1}{2} - 2\delta \frac{e^v + u}{1+u} |\nabla \log v|^2 \right] \nabla \log v \right|^2 \\ - \frac{vu}{4(e^v + u)} |\nabla \log v|^2 + 2\delta \frac{vu}{1+u} |\nabla \log v|^4 - 4\delta^2 v \frac{u(e^v + u)}{(1+u)^2} |\nabla \log v|^6. \end{aligned} \quad (2.42)$$

Direct maximization in $|\nabla \log v|^2$, together with (2.38), gives

$$\begin{aligned} \int_{B_R} \left[\frac{vu}{4(e^v + u)} |\nabla \log v|^2 - 2\delta \frac{vu}{1+u} |\nabla \log v|^4 + 4\delta^2 v \frac{u(e^v + u)}{(1+u)^2} |\nabla \log v|^6 \right] \\ \leq \frac{1}{128\delta} \int_{B_R} uv + 4\Lambda_0 \delta^2 \int_{B_R} \frac{|\nabla v|^6}{v^5}. \end{aligned} \quad (2.43)$$

The same completion and the quadratic triangle inequality prove (2.37).

Take $\alpha = 1$ and $\beta = 3/56$ in (2.41). The sharp inequality (2.28) gives

$$\begin{aligned} 3 \left[\int_{B_R} v |\nabla \log v|^2 |D^2 \log v|^2 + 2 \int_{B_R} v |D^2 \log v \nabla \log v|^2 \right] - \frac{3}{56} \int_{B_R} \frac{|\nabla v|^6}{v^5} \\ \geq \left[\int_{B_R} v |\nabla \log v|^2 |D^2 \log v|^2 + 2 \int_{B_R} v |D^2 \log v \nabla \log v|^2 \right] + \frac{9}{56} \int_{B_R} \frac{|\nabla v|^6}{v^5}. \end{aligned}$$

Consequently,

$$\begin{aligned}
& \frac{d}{dt}(\mathfrak{e} + \delta \mathfrak{q}) + \int_{B_R} v \frac{u}{e^v + u} \left| \nabla u - u \nabla v - \left[\frac{1}{2} - 2\delta \frac{e^v + u}{1 + u} |\nabla \log v|^2 \right] \nabla \log v \right|^2 \\
& + \delta \left[\int_{B_R} v |\nabla \log v|^2 |D^2 \log v|^2 + 2 \int_{B_R} v |D^2 \log v \nabla \log v|^2 \right] + \frac{9\delta}{56} \int_{B_R} \frac{|\nabla v|^6}{v^5} \\
& + \delta \int_{B_R} uv |\nabla \log v|^4 \\
& \leq \int_{B_R} \left[\frac{vu}{4(e^v + u)} |\nabla \log v|^2 - 2\delta \frac{vu}{1 + u} |\nabla \log v|^4 + 4\delta^2 v \frac{u(e^v + u)}{(1 + u)^2} |\nabla \log v|^6 \right] \\
& + (\|v_0\|_{L^\infty} + 16\delta) \int_{B_R} \frac{u}{v} |\nabla v|^2 + \int_{B_R} u^2 v + C_\delta \int_{B_R} uv.
\end{aligned} \tag{2.44}$$

Use Lemma 2.10 with $\varepsilon = 1/4$ and $\eta = \delta/4$, followed by (2.37). Also use (2.33) with

$$\sigma = \frac{\delta}{4(\|v_0\|_{L^\infty} + 16\delta)}.$$

Together with (2.43), these choices retain one half of the completed square, the full logarithmic Hessian form, one half of the reaction score, and the sixth-order coefficient $9\delta/56 - 6\Lambda_0\delta^2$. This coefficient is at least $9\delta/112$ when

$$0 < \delta \leq \min \left\{ 1, \frac{3}{224\Lambda_0} \right\}.$$

Dropping the nonnegative boundary term proves (2.36). $\square$

For later recovery, the Fisher identity and convexity of B_R give

$$\frac{d}{dt} \int_{B_R} \frac{|\nabla v|^2}{v} + 2 \int_{B_R} v |D^2 \log v|^2 + \int_{B_R} \frac{u}{v} |\nabla v|^2 \leq -2 \int_{B_R} \nabla u \cdot \nabla v.$$

Using

$$\nabla u = \nabla \log(1 + u) + \frac{u}{1 + u} \nabla u,$$

we obtain from (2.27) and (2.29)

$$2 \left| \int_{B_R} \log(1 + u) \Delta v \right| \leq \int_{B_R} v |D^2 \log v|^2 + C \int_{B_R} uv,$$

and

$$2 \int_{B_R} \frac{u}{1 + u} |\nabla u| |\nabla v| \leq 4 \int_{B_R} \frac{uv}{(1 + u)^2} |\nabla u|^2 + \frac{1}{4} \int_{B_R} \frac{u}{v} |\nabla v|^2.$$

Consequently,

$$\begin{aligned}
& \frac{d}{dt} \int_{B_R} \frac{|\nabla v|^2}{v} + \int_{B_R} v |D^2 \log v|^2 + \frac{3}{4} \int_{B_R} \frac{u}{v} |\nabla v|^2 \\
& \leq 4 \int_{B_R} \frac{uv}{(1 + u)^2} |\nabla u|^2 + C \int_{B_R} uv.
\end{aligned} \tag{2.45}$$

Proposition 2.12. *There is $\delta > 0$, independent of R and T , such that every positive classical solution of (2.22) satisfies*

$$\sup_{0 < t < T} \left[\int_{B_R} e^v [(1 + ue^{-v}) \log(1 + ue^{-v}) - ue^{-v}] + \delta \int_{B_R} \frac{|\nabla v|^4}{v^3} \right] \leq C, \tag{2.46}$$

and

$$\begin{aligned}
& \int_0^T \int_{B_R} \left[v \frac{ue^{-v}}{1 + ue^{-v}} |\nabla u - u \nabla v|^2 + v |\nabla \log v|^2 |D^2 \log v|^2 + 2v |D^2 \log v \nabla \log v|^2 \right. \\
& \left. + \frac{|\nabla v|^6}{v^5} + uv |\nabla \log v|^4 + uv \right] \leq C.
\end{aligned} \tag{2.47}$$

The following estimates are consequences of this two-multiplier bound:

$$\sup_{0 < t < T} \left[\int_{B_R} |\nabla v|^2 + \int_{B_R} \frac{|\nabla v|^2}{v} \right] \leq C, \quad (2.48)$$

$$\int_0^T \int_{B_R} \left[\frac{uv}{(1+u)^2} |\nabla u|^2 + |\Delta v|^2 + v |D^2 \log v|^2 + \frac{u}{v} |\nabla v|^2 + u^2 v + uv^2 \right] \leq C. \quad (2.49)$$

The constants are independent of R and T .

Proof. Add $(C_\delta + 1)$ times the nutrient mass identity to (2.36), and integrate in time. The initial chemical energy is uniformly bounded because

$$(1+s) \log(1+s) - s \leq s \log(1+s), \quad s \geq 0,$$

and hence

$$e^{v_{0,R}} [(1 + u_{0,R} e^{-v_{0,R}}) \log(1 + u_{0,R} e^{-v_{0,R}}) - u_{0,R} e^{-v_{0,R}}] \leq u_{0,R} \log(1 + u_{0,R}).$$

Lemma 2.6 and (2.19) control the remaining terms at time zero. The recovery estimate (2.37) then proves (2.46) and (2.47).

It remains to recover the auxiliary estimates used in the compactness argument. The pointwise inequality

$$\int_{B_R} \frac{u}{v} |\nabla v|^2 \leq \int_{B_R} uv |\nabla \log v|^4 + \frac{1}{4} \int_{B_R} uv$$

gives the required time integral. Lemma 2.10 proves the $u^2 v$ -part of (2.49). Moreover,

$$\begin{aligned} \int_{B_R} \frac{uv}{(1+u)^2} |\nabla u|^2 &\leq 2 \int_{B_R} \frac{uv}{(1+u)^2} |\nabla u - u \nabla v|^2 + 2 \int_{B_R} \frac{u^3 v}{(1+u)^2} |\nabla v|^2 \\ &\leq C \int_{B_R} v \frac{u}{e^v + u} |\nabla u - u \nabla v|^2 + 2 \|v_0\|_{L^\infty}^2 \int_{B_R} \frac{u}{v} |\nabla v|^2. \end{aligned}$$

Testing the nutrient equation by $-\Delta v$ gives

$$\frac{1}{2} \frac{d}{dt} \int_{B_R} |\nabla v|^2 + \frac{1}{2} \int_{B_R} |\Delta v|^2 \leq \frac{1}{2} \int_{B_R} u^2 v^2 \leq \frac{\|v_0\|_{L^\infty}}{2} \int_{B_R} u^2 v.$$

The initial bound (2.21) therefore yields the Dirichlet part of (2.48) and the Δv -part of (2.49).

The Fisher information obeys the algebraic estimate

$$\left(\int_{B_R} \frac{|\nabla v|^2}{v} \right)^2 \leq \left(\int_{B_R} v \right) \mathfrak{q}.$$

Its differential inequality (2.45), together with the recovered population gradient and the preceding reaction-score estimate, gives $\int_0^T \int_{B_R} v |D^2 \log v|^2 \leq C$. Finally, $uv^2 \leq \|v_0\|_{L^\infty} uv$. This completes (2.48) and (2.49). $\square$

3. UNIFORM BOUNDEDNESS AND THE WHOLE-SPACE LIMIT

3.1. Interpolation and uniform population bounds. The next lemma is centered at the parent exponent r . The sixth logarithmic-gradient moment lifts the parent moment r to the child energy exponent $3r - 1$, because the endpoint estimate applied to $f^{3r/2} \sqrt{g}$ contains the exact factorization

$$f^{3r/2} g^{-1/2} |\nabla g| = \left(\frac{|\nabla g|^6}{g^5} \right)^{1/6} (f^{3r} g)^{1/3} (f^r)^{1/2}.$$

Thus Hölder's inequality with exponents 6, 3, 2 leaves precisely the parent moment $\int f^r$.

Lemma 3.1. *There is $C > 0$ such that, for every $R \geq 1$, $r \geq 1$, $0 < \eta \leq 1$, and positive $f, g \in C^1(\overline{B_R})$, one has*

$$\begin{aligned} \int_{B_R} f^{3r} g &\leq \eta \int_{B_R} f^{3r-2} g |\nabla f|^2 \\ &\quad + C \left(\int_{B_R} f^r \right)^3 \int_{B_R} \frac{|\nabla g|^6}{g^5} \end{aligned}$$

$$+ Cr^{6-2/r} \eta^{-(3-1/r)} \left(\int_{B_R} f^r \right)^{3-1/r} \int_{B_R} fg. \quad (3.1)$$

Proof. Fix $\ell > 0$, and write

$$\omega_\ell(f) = \left(\frac{f^r}{f^r + \ell^r} \right)^{1/2}.$$

Splitting into $f \leq \ell$ and $f > \ell$ gives

$$f^{3r}g \leq \ell^{3r-1}fg + 2f^{3r}\omega_\ell(f)^2g. \quad (3.2)$$

Apply Lemma 2.8 with

$$q_r = \frac{2r}{4r-1} \in \left[\frac{1}{2}, \frac{2}{3} \right]$$

to $f^{3r/2}\omega_\ell(f)\sqrt{g}$. Direct differentiation yields

$$\left| \nabla \left(f^{3r/2}\omega_\ell(f) \right) \right| \leq 2rf^{(3r-2)/2}\omega_\ell(f)|\nabla f|.$$

Moreover,

$$\omega_\ell(f)^2 \leq \ell^{-r}f^r.$$

The f -derivative, the g -derivative, and the lower endpoint term are therefore controlled by

$$\begin{aligned} \left(\int_{B_R} f^{(3r-2)/2} \sqrt{g} \omega_\ell(f) |\nabla f| \right)^2 &\leq \ell^{-r} \left(\int_{B_R} f^{3r-2} g |\nabla f|^2 \right) \left(\int_{B_R} f^r \right), \\ \int_{B_R} f^{3r/2} \omega_\ell(f) g^{-1/2} |\nabla g| &\leq \left(\int_{B_R} \frac{|\nabla g|^6}{g^5} \right)^{1/6} \left(\int_{B_R} f^{3r} g \right)^{1/3} \left(\int_{B_R} f^r \right)^{1/2}, \\ \left(\int_{B_R} \left[f^{3r/2} \omega_\ell(f) \sqrt{g} \right]^{q_r} \right)^{2/q_r} &\leq \left(\int_{B_R} f^r \right)^{3-1/r} \int_{B_R} fg. \end{aligned}$$

The second bound follows from the factorization preceding the lemma and Hölder's inequality with exponents 6, 3, 2. For the last bound, observe directly that

$$\left[f^{3r/2} \omega_\ell(f) \sqrt{g} \right]^{q_r} \leq (f^r)^{(3r-1)/(4r-1)} (fg)^{r/(4r-1)}.$$

The two exponents add to one, and their ratio is $3 - 1/r$. Hölder's inequality gives the last bound.

Combining these bounds with (3.2), applying the endpoint inequality, and absorbing one half of $\int_{B_R} f^{3r}g$ gives the pre-optimized estimate

$$\begin{aligned} \int_{B_R} f^{3r}g &\leq Cr^2 \ell^{-r} \left(\int_{B_R} f^{3r-2} g |\nabla f|^2 \right) \left(\int_{B_R} f^r \right) \\ &\quad + C \left(\int_{B_R} f^r \right)^3 \int_{B_R} \frac{|\nabla g|^6}{g^5} \\ &\quad + C \left[\ell^{3r-1} + R^{-2(3-1/r)} \left(\int_{B_R} f^r \right)^{3-1/r} \right] \int_{B_R} fg. \end{aligned} \quad (3.3)$$

Here the only nonlinear absorption generated by the derivative of g uses the sharp scalar inequality

$$AB^{2/3} \leq \varepsilon B + \frac{4}{27\varepsilon^2} A^3, \quad A, B, \varepsilon > 0.$$

Choose

$$\ell^r = Cr^2 \eta^{-1} \int_{B_R} f^r,$$

where the fixed constant is sufficiently large. The first term in (3.3) is absorbed into the weighted population gradient. Moreover,

$$\ell^{3r-1} \leq Cr^{6-2/r} \eta^{-(3-1/r)} \left(\int_{B_R} f^r \right)^{3-1/r},$$

and $R \geq 1$, $r \geq 1$, and $\eta \leq 1$ absorb the radius term into the same lower-order contribution. This proves (3.1). $\square$

Proposition 3.2. *There is $C > 0$, independent of R and T , such that*

$$\sup_{0 < t < T} \|u(t)\|_{L^\infty(B_R)} \leq C, \quad (3.4)$$

and

$$\int_0^T \int_{B_R} (uv|\nabla u|^2 + u^3v|\nabla v|^2) \leq C.$$

Proof. Proposition 2.12 gives

$$\sup_{0 < t < T} \int_{B_R} \frac{|\nabla v|^4}{v^3} \leq C.$$

For $p \geq 2$, applying Lemma 2.8 with $q = 1$ to $u^{p+1}v^{5/2}$, followed by Young's inequality, yields, for $0 < \gamma \leq 1$,

$$\int_{B_R} u^{p+1}v|\nabla v|^2 \leq \gamma \int_{B_R} u^{p-1}v|\nabla u|^2 + C_\gamma p^2 \int_{B_R} u^{p+1}v^2. \quad (3.5)$$

Here the quartic bound first gives

$$\int_{B_R} u^{p+1}v|\nabla v|^2 \leq C \|u^{p+1}v^{5/2}\|_{L^2(B_R)}.$$

The two derivative terms in the endpoint estimate are bounded by

$$Cp \left(\int_{B_R} u^{p-1}v|\nabla u|^2 \right)^{1/2} \left(\int_{B_R} u^{p+1}v^2 \right)^{1/2}$$

and

$$C \left(\int_{B_R} u^{p+1}v|\nabla v|^2 \right)^{1/2} \left(\int_{B_R} u^{p+1}v^2 \right)^{1/2},$$

while the scaled lower-order term is controlled by $C \int_{B_R} u^{p+1}v^2$.

Choosing γ sufficiently small, testing the population equation by pu^{p-1} , and using (3.5) give, for some $c > 0$,

$$\frac{d}{dt} \int_{B_R} u^p + cp^2 \int_{B_R} u^{p-1}v|\nabla u|^2 \leq Cp^4 \int_{B_R} u^{p+1}v^2 + p \int_{B_R} u^p v. \quad (3.6)$$

Set $r = (p+1)/3$. Apply Lemma 3.1 at this parent exponent first with $g = v^2$ and η chosen as a sufficiently small fixed multiple of p^{-2} , and then with $g = v$ and a fixed sufficiently small η . Since

$$\frac{|\nabla(v^2)|^6}{(v^2)^5} \leq 64 \|v_0\|_{L^\infty} \frac{|\nabla v|^6}{v^5}, \quad \int_{B_R} uv^2 \leq \|v_0\|_{L^\infty} \int_{B_R} uv,$$

the two applications give

$$Cp^4 \int_{B_R} u^{p+1}v^2 \leq \frac{c}{4} p^2 \int_{B_R} u^{p-1}v|\nabla u|^2 + Cp^4 \left(\int_{B_R} u^r \right)^3 \int_{B_R} \frac{|\nabla v|^6}{v^5} + Cp^{16} \left(1 + \int_{B_R} u^r \right)^3 \int_{B_R} uv, \quad (3.7)$$

$$p \int_{B_R} u^p v \leq \frac{c}{4} p^2 \int_{B_R} u^{p-1}v|\nabla u|^2 + Cp \left(\int_{B_R} u^r \right)^3 \int_{B_R} \frac{|\nabla v|^6}{v^5} + Cp^7 \left(1 + \int_{B_R} u^r \right)^3 \int_{B_R} uv. \quad (3.8)$$

Here we used $u^p \leq u + u^{p+1}$, $p/r = 3p/(p+1) < 3$, and enlarged the powers of p to fixed polynomial bounds. Substitution into (3.6) yields the single recursion inequality

$$\frac{d}{dt} \int_{B_R} u^p + cp^2 \int_{B_R} u^{p-1}v|\nabla u|^2 \leq Cp^{16} \left(1 + \int_{B_R} u^r \right)^3 \left(\int_{B_R} \frac{|\nabla v|^6}{v^5} + \int_{B_R} uv \right), \quad p \geq 2. \quad (3.9)$$

After increasing C_0 once, time integration, Proposition 2.12, and (2.20) imply

$$1 + \sup_{0 < t < T} \int_{B_R} u^p(t) \leq C_0^p + Cp^{16} \left(1 + \sup_{0 < t < T} \int_{B_R} u^{(p+1)/3}(t) \right)^3.$$

The exponent lift in this inequality is affine and has fixed point $1/2$. Set

$$p_k = \frac{3^k + 1}{2}, \quad x_k = \max \left\{ C_0, \left(1 + \sup_{0 < t < T} \int_{B_R} u^{p_k}(t) \right)^{2/3^k} \right\}, \quad k \geq 0.$$

The mass identity controls x_0 . Since $(p_k + 1)/3 = p_{k-1}$, the preceding estimate at $p = p_k$, with $k \geq 1$, gives

$$x_k \leq \left(C_0^{1/2} + Cp_k^{16}\right)^{2/3^k} x_{k-1}. \quad (3.10)$$

Since

$$\sum_{k=1}^{\infty} \frac{2}{3^k} \log(C_0^{1/2} + Cp_k^{16}) < \infty,$$

iteration of (3.10) yields $\sup_k x_k < \infty$, uniformly in R and T . Moreover,

$$\sup_{0 < t < T} \|u(t)\|_{L^{p_k}(B_R)} \leq x_k^{3^k/(2p_k)}.$$

Because $3^k/(2p_k) \rightarrow 1$, letting $k \rightarrow \infty$ proves (3.4).

Finally, take $p = 2$ in (3.9) before dropping its gradient term and integrate in time. This gives

$$\int_0^T \int_{B_R} uv |\nabla u|^2 \leq C.$$

Using (3.5) at $p = 2$, together with Lemma 3.1 for $f = u$, $g = v^2$, and $r = 1$, gives

$$\int_0^T \int_{B_R} u^3 v |\nabla v|^2 \leq C.$$

□

3.2. Continuation and local estimates.

Corollary 3.3. *For each $R = R_j$ and the corresponding prepared initial data, $T_{\max, R} = \infty$. Uniformly in R ,*

$$\begin{aligned} \int_0^\infty \int_{B_R} \left(|uv \nabla u|^2 + |u^2 v \nabla v|^2 + (uv)^2 + |\Delta v|^2 \right) &\leq C, \\ \int_0^\infty \int_{B_R} \left(|uv \nabla u| + |u^2 v \nabla v| + uv \right) &\leq C. \end{aligned} \quad (3.11)$$

Proof. Proposition 3.2 excludes the extensibility alternative (2.23). The L^2 -flux estimates follow from

$$|uv \nabla u|^2 = (uv) uv |\nabla u|^2, \quad |u^2 v \nabla v|^2 = (uv) u^3 v |\nabla v|^2, \quad (uv)^2 \leq Cuv,$$

together with Propositions 2.12 and 3.2. The Cauchy–Schwarz inequality and the uniform bound for $\int_0^\infty \int_{B_R} uv$ then yield (3.11). □

Return to the sequence R_j from Lemma 2.6, and denote the corresponding global solutions by (u_j, v_j) .

Lemma 3.4. *For every compact $K \Subset \mathbb{R}^2$ and $0 < \tau < T < \infty$, there are j_0 and $c_{K, \tau, T} > 0$ such that*

$$v_j \geq c_{K, \tau, T} \quad \text{on } K \times [\tau, T]$$

for every $j \geq j_0$.

Proof. Choose $r > 0$ with $\int_{B_r} v_0 > 0$, and choose a bounded smooth connected domain Ω_1 with $K \cup \overline{B_r} \Subset \Omega_1$. For large j , $\Omega_1 \Subset B_{R_j}$ and

$$\int_{B_r} v_{0,j} \geq \frac{1}{2} \int_{B_r} v_0.$$

Proposition 3.2 gives $u_j \leq C$. Let k_{Ω_1} be the Dirichlet heat kernel on Ω_1 . The comparison principle for $(v_j)_t \geq \Delta v_j - Cv_j$, followed by comparison of the restricted Neumann flow with the Dirichlet flow on Ω_1 , yields, for $(x, t) \in K \times [\tau, T]$,

$$v_j(x, t) \geq e^{-Ct} e^{t\Delta_{N, B_{R_j}}} v_{0,j}(x) \geq e^{-Ct} \int_{B_r} k_{\Omega_1}(t, x, y) v_{0,j}(y) dy \geq c_{K, \tau, T}.$$

For the second comparison, the Dirichlet solution extended by zero is a subsolution in B_{R_j} . The last constant is positive because k_{Ω_1} has a positive minimum on $[\tau, T] \times K \times \overline{B_r}$. □

Lemma 3.5. *Let $\chi \in C_c^\infty(\mathbb{R}^2; [0, 1])$. For all sufficiently large j ,*

$$\int_0^\infty \int_{B_{R_j}} \chi^2 v_j |\nabla u_j|^2 \leq C_\chi. \quad (3.12)$$

Consequently, for every compact $K \Subset \mathbb{R}^2$ and $0 < \tau < T$,

$$\sup_{j \geq j_0} \int_\tau^T \int_K |\nabla u_j|^2 < \infty. \quad (3.13)$$

Proof. For large j , $\text{supp } \chi \Subset B_{R_j}$. Testing the population equation by $\chi^2 \log u_j$ gives

$$\begin{aligned} \frac{d}{dt} \int_{B_{R_j}} \chi^2 (u_j \log u_j - u_j + 1) + \int_{B_{R_j}} \chi^2 v_j |\nabla u_j|^2 &= \int_{B_{R_j}} \chi^2 u_j v_j \nabla u_j \cdot \nabla v_j \\ &\quad - 2 \int_{B_{R_j}} \chi u_j v_j \log u_j \nabla u_j \cdot \nabla \chi \\ &\quad + 2 \int_{B_{R_j}} \chi u_j^2 v_j \log u_j \nabla v_j \cdot \nabla \chi \\ &\quad + \int_{B_{R_j}} \chi^2 u_j v_j \log u_j. \end{aligned}$$

The uniform bounds $0 < v_j \leq \|v_0\|_{L^\infty}$ and $0 < u_j \leq C$, together with $u_j v_j^2 \leq C$, imply

$$s^2 \log^2 s \leq Cs, \quad s^3 \log^2 s \leq Cs, \quad s \log s \leq Cs \quad \text{for } 0 < s \leq \sup_j \|u_j\|_\infty.$$

Applying Young's inequality to the four terms on the right of the testing identity gives, using these three scalar bounds,

$$\frac{d}{dt} \int_{B_{R_j}} \chi^2 (u_j \log u_j - u_j + 1) + \frac{1}{2} \int_{B_{R_j}} \chi^2 v_j |\nabla u_j|^2 \leq C_\chi \left(\int_{B_{R_j}} \frac{u_j}{v_j} |\nabla v_j|^2 + \int_{B_{R_j}} u_j v_j \right).$$

The initial localized entropy is uniformly bounded because the prepared population data are uniformly bounded. Proposition 2.12 and (2.24) make the right-hand side integrable on $(0, \infty)$, which proves (3.12). Choose $\chi = 1$ on a neighborhood of K , and use Lemma 3.4 on $K \times [\tau, T]$ to obtain (3.13). $\square$

3.3. Exhaustion compactness and passage to the whole space.

Proposition 3.6. *There exist a subsequence, still indexed by j , and nonnegative functions u, v such that, for every bounded smooth $\Omega \Subset \mathbb{R}^2$, every $T > 0$, and every $0 < \tau < T$,*

$$v_j \rightarrow v \quad \text{strongly in } L^2(0, T; H^1(\Omega)), \quad (3.14)$$

$$u_j \rightarrow u \quad \text{strongly in } L^p(\Omega \times (\tau, T)) \quad \text{for every } 1 \leq p < \infty,$$

$$\nabla u_j \rightharpoonup \nabla u \quad \text{weakly in } L^2(\Omega \times (\tau, T)). \quad (3.15)$$

The weighted nutrient variables satisfy

$$v_j^{1/4} \xrightarrow{*} v^{1/4} \quad \text{in } L^\infty(0, T; W^{1,4}(\Omega)), \quad (3.16)$$

$$v_j^{1/6} \rightharpoonup v^{1/6} \quad \text{in } L^6(0, T; W^{1,6}(\Omega)). \quad (3.17)$$

After zero extension outside B_{R_j} ,

$$u_j v_j \nabla u_j \rightharpoonup uv \nabla u, \quad (3.18)$$

$$u_j^2 v_j \nabla v_j \rightharpoonup u^2 v \nabla v, \quad (3.19)$$

$$\sqrt{u_j v_j} \nabla u_j \rightharpoonup \sqrt{uv} \nabla u, \quad (3.20)$$

$$u_j^{3/2} \sqrt{v_j} \nabla v_j \rightharpoonup u^{3/2} \sqrt{v} \nabla v \quad (3.21)$$

weakly in $L^2(\mathbb{R}^2 \times (0, \infty); \mathbb{R}^2)$. In addition, after zero extension,

$$\Delta v_j \rightharpoonup \Delta v \quad \text{weakly in } L^2(\mathbb{R}^2 \times (0, \infty)). \quad (3.22)$$

Proof. Fix smooth domains $\Omega_0 \Subset \Omega_1 \Subset \Omega_2 \Subset \mathbb{R}^2$. Corollary 3.3 gives uniform L^2 -bounds for Δv_j and $u_j v_j$. The mass and pointwise bounds control v_j in L^2 . Interior elliptic estimates on $\Omega_1 \Subset \Omega_2$ therefore give

$$\sup_j \|v_j\|_{L^2(0,T;H^2(\Omega_1))} < \infty,$$

and the nutrient equation gives

$$\sup_j \|\partial_t v_j\|_{L^2(\Omega_1 \times (0,T))} < \infty.$$

Aubin–Lions yields (3.14) on Ω_0 , after restriction and a diagonal extraction.

Lemma 3.5, with a cutoff equal to one on Ω_1 and supported in Ω_2 , and the uniform population bound give

$$\sup_j \|u_j\|_{L^2(\tau,T;H^1(\Omega_1))} < \infty.$$

By Corollary 3.3,

$$\partial_t u_j = \nabla \cdot (u_j v_j \nabla u_j - u_j^2 v_j \nabla v_j) + u_j v_j$$

is bounded in $L^2(\tau,T;H^{-1}(\Omega_1))$. Hence

$$H^1(\Omega_1) \Subset L^2(\Omega_1) \hookrightarrow H^{-1}(\Omega_1)$$

and the Aubin–Lions lemma give strong L^2 -compactness of u_j on $\Omega_0 \times (\tau,T)$. The uniform L^∞ -bound upgrades this convergence to every finite L^p , and weak compactness gives (3.15).

The fluxes are bounded in $L^2(\mathbb{R}^2 \times (0,\infty);\mathbb{R}^2)$ by Corollary 3.3. On a fixed positive-time compact cylinder, $u_j v_j \rightarrow uv$ and $u_j^2 v_j \rightarrow u^2 v$ strongly in every finite L^p -space. Pairing with $\nabla u_j \rightharpoonup \nabla u$ and using $\nabla v_j \rightarrow \nabla v$ strongly in L^2 identifies the distributional limits of the two fluxes. Their global weak L^2 -limits are therefore the expressions in (3.18) and (3.19).

The entropy bounds also give, on every fixed Ω ,

$$\sup_j \|v_j^{1/4}\|_{L^\infty(0,T;W^{1,4}(\Omega))} < \infty, \quad \sup_j \|v_j^{1/6}\|_{L^6(0,T;W^{1,6}(\Omega))} < \infty.$$

Indeed, the gradient terms follow from

$$\frac{|\nabla v_j|^4}{v_j^3} = 256 |\nabla v_j^{1/4}|^4, \quad \frac{|\nabla v_j|^6}{v_j^5} = 6^6 |\nabla v_j^{1/6}|^6,$$

whereas the zero-order terms follow from the mass bound. After a further diagonal extraction, weak and weak-star compactness give candidate limits in the spaces stated in (3.16) and (3.17). The strong convergence (3.14) yields almost-everywhere convergence on compact cylinders. It therefore identifies these limits as $v^{1/4}$ and $v^{1/6}$.

Finally, Proposition 3.2 and Corollary 3.3 bound the two sequences on the left of (3.20) and (3.21) in $L^2(\mathbb{R}^2 \times (0,\infty);\mathbb{R}^2)$. On each positive-time compact cylinder, the scalar coefficients converge strongly in every finite L^p -space. The weak convergence of ∇u_j and the strong convergence of ∇v_j identify the distributional limits. Global weak compactness then gives (3.20) and (3.21). The local distributional convergence of v_j similarly identifies the global weak L^2 -limit of the zero extensions of Δv_j as Δv , proving (3.22). $\square$

Lemma 3.7. *The limit nutrient has an L^1 -continuous representative satisfying*

$$v(\cdot, t) = e^{t\Delta} v_0 - \int_0^t e^{(t-s)\Delta} (uv)(\cdot, s) \, ds \quad \text{in } L^1(\mathbb{R}^2), \quad (3.23)$$

and

$$0 \leq v(\cdot, t) \leq e^{t\Delta} v_0 \quad \text{a.e.} \quad (3.24)$$

The same representative satisfies

$$\operatorname{ess\,sup}_{t>0} \|\nabla v(\cdot, t)\|_{L^\infty(\mathbb{R}^2)} \leq C. \quad (3.25)$$

Proof. We first pass to the limit in the ball nutrient identities. Proposition 3.6 handles every positive-time compact cylinder. On $(0, t_0)$, the diffusion contribution is controlled by the uniform bound for $\int_{B_{R_j}} |\nabla v_j|^2$, the reaction contribution by the pointwise bounds for u_j and v_j , and the initial contribution by the convergence $v_{0,j} \rightarrow v_0$ in $L^1(\mathbb{R}^2)$. With a fixed compactly supported test function, first let $j \rightarrow \infty$, and then let $t_0 \downarrow 0$. Thus v satisfies the weak nutrient equation with initial value v_0 . The ball consumption identity, local almost-everywhere convergence, a countable spatial exhaustion, and Fatou's lemma give $uv \in L^1(\mathbb{R}^2 \times (0, T))$. Uniqueness of the L^1 -mild solution of the inhomogeneous heat equation gives (3.23). Positivity of the heat semigroup gives (3.24). For $0 < t \leq 1$,

$$\|\nabla v(t)\|_\infty \leq \|\nabla v_0\|_\infty + C \int_0^t (t-s)^{-1/2} \|u(s)v(s)\|_\infty ds \leq C.$$

For $t \geq 1$, apply the same formula on $[t-1, t]$:

$$\|\nabla v(t)\|_\infty \leq C\|v(t-1)\|_\infty + C \int_{t-1}^t (t-s)^{-1/2} \|u(s)v(s)\|_\infty ds. \quad (3.26)$$

The uniform bounds for u and v now give (3.25). $\square$

Lemma 3.8 (Population representative). *Suppose that a nonnegative function u satisfies the population equation in distributions with initial value u_0 , that $uv\nabla u - u^2v\nabla v$ and uv belong to $L^1(\mathbb{R}^2 \times (0, \infty))$, and that*

$$\operatorname{ess\,sup}_{0 < t < T} (\|u(t)\|_{L^1} + \|u(t)\|_{L^\infty}) < \infty \quad \text{for every } T > 0.$$

Then u admits a representative in $C([0, \infty); \operatorname{BL}^(\mathbb{R}^2))$. Each time slice of this representative is a nonnegative function in $L^1(\mathbb{R}^2) \cap L^\infty(\mathbb{R}^2)$, with the same bounds as the almost-everywhere time slices. Furthermore, for every $0 \leq t_1 < t_2$ and $\varphi \in W^{1,\infty}(\mathbb{R}^2)$,*

$$\int_{\mathbb{R}^2} (u(t_2) - u(t_1))\varphi = - \int_{t_1}^{t_2} \int_{\mathbb{R}^2} (uv\nabla u - u^2v\nabla v) \cdot \nabla \varphi + \int_{t_1}^{t_2} \int_{\mathbb{R}^2} uv\varphi. \quad (3.27)$$

In particular,

$$\int_{\mathbb{R}^2} u(t) = \int_{\mathbb{R}^2} u_0 + \int_0^t \int_{\mathbb{R}^2} uv \quad \text{for every } t \geq 0. \quad (3.28)$$

Moreover, for every $R \geq 1$,

$$\sup_{t \geq 0} \int_{\mathbb{R}^2 \setminus B_{2R}} u(t) \leq \int_{\mathbb{R}^2 \setminus B_R} u_0 + \frac{C}{R} \int_0^\infty \int_{\mathbb{R}^2} |uv\nabla u - u^2v\nabla v| + \int_0^\infty \int_{\mathbb{R}^2 \setminus B_R} uv. \quad (3.29)$$

Proof. For $\varphi \in C_c^\infty(\mathbb{R}^2)$, the distributional equation shows that $t \mapsto \int u(t)\varphi$ has an absolutely continuous representative whose increment is the right-hand side of (3.27). Choose a countable family of smooth compactly supported test functions that is dense in the test-function topology. Outside one common null set of times, the identity holds for this family and hence for every such test function. If φ is compactly supported and Lipschitz, choose mollifications with support in one fixed compact set such that

$$\varphi_m \rightarrow \varphi \quad \text{uniformly}, \quad \nabla \varphi_m \rightarrow \nabla \varphi \quad \text{a.e.}, \quad \sup_m \|\nabla \varphi_m\|_\infty \leq \|\nabla \varphi\|_\infty.$$

Dominated convergence for the three terms in the population equation extends the identity to these test functions.

For every $t \geq 0$, define the representative by

$$\langle u(t), \varphi \rangle = \int_{\mathbb{R}^2} u_0 \varphi - \int_0^t \int_{\mathbb{R}^2} (uv\nabla u - u^2v\nabla v) \cdot \nabla \varphi + \int_0^t \int_{\mathbb{R}^2} uv\varphi, \quad \varphi \in W^{1,\infty}(\mathbb{R}^2).$$

Spatial cutoffs show that this representative agrees with the original almost-everywhere time slices outside the common null set. The definition gives

$$\|u(t_2) - u(t_1)\|_{\operatorname{BL}^*} \leq \int_{t_1}^{t_2} \int_{\mathbb{R}^2} (|uv\nabla u - u^2v\nabla v| + uv). \quad (3.30)$$

Until the density property is proved, the left-hand side denotes the operator norm on bounded Lipschitz test functions.

Let $\chi_R \in C_c^\infty(\mathbb{R}^2; [0, 1])$ equal one on B_R , vanish outside B_{2R} , and satisfy $|\nabla \chi_R| \leq C/R$. Test the identity from time zero with $1 - \chi_R$, first multiplied by a compactly supported outer cutoff, and then remove that cutoff by dominated convergence. At every Lebesgue time this gives

$$\int_{\mathbb{R}^2 \setminus B_{2R}} u(t) \leq \int_{\mathbb{R}^2 \setminus B_R} u_0 + \frac{C}{R} \int_0^t \int_{\mathbb{R}^2} |uv \nabla u - u^2 v \nabla v| + \int_0^t \int_{\mathbb{R}^2 \setminus B_R} uv.$$

Extending the time integrals to infinity proves the uniform tail estimate at all Lebesgue times.

Fix $t \geq 0$ and choose Lebesgue times $t_n \rightarrow t$ outside the common null set. The functions $u(t_n)$ are uniformly bounded in $L^1 \cap L^\infty$, and the preceding estimate makes them tight. Their L^∞ -bound also gives uniform integrability on sets of small measure. The Dunford–Pettis theorem therefore provides a subsequence converging weakly in L^1 to a nonnegative function with the same L^∞ -bound. By (3.30), $u(t_n)$ converges to the functional $u(t)$ on every bounded Lipschitz test function. Thus the weak L^1 -limit is the density of $u(t)$. It is independent of the subsequence and the Lebesgue times. This proves the asserted continuity, (3.27), and (3.29) at every time. Taking $\varphi \equiv 1$ gives (3.28); taking $t_1 = 0$ in (3.30) gives the initial trace. $\square$

Proof of Theorem 2.2 and Proposition 2.3. Lemma 3.7 already supplies the nutrient weak equation and its initial value. To pass to the limit in the population equation, fix $T > 0$ and use Proposition 3.6 on positive-time compact cylinders. The L^2 -flux bounds give

$$\sup_j \int_0^{t_0} \int_K (|u_j v_j \nabla u_j| + |u_j^2 v_j \nabla v_j|) \leq C_K t_0^{1/2},$$

and the mass and pointwise bounds give an $O(t_0)$ contribution from u_j and $u_j v_j$. Letting $t_0 \downarrow 0$ and using the local convergence of the prepared population data proves (2.4).

The local positivity estimate in Lemma 3.4 passes to the limit and proves (2.6). Lemma 3.5 and weak lower semicontinuity give (2.3). The population equation and the weak L^2 -limits of the two fluxes give the stated H^{-1} -regularity of u_t .

For every bounded $\Omega \Subset \mathbb{R}^2$ and $T > 0$, the weighted lower semicontinuity can be written directly in the root variables. Equations (3.16) and (3.17) give

$$\begin{aligned} 256 \operatorname{ess\,sup}_{0 < t < T} \int_\Omega |\nabla v^{1/4}(t)|^4 &\leq \liminf_{j \rightarrow \infty} \operatorname{ess\,sup}_{0 < t < T} \int_{B_{R_j}} \frac{|\nabla v_j|^4}{v_j^3}, \\ 6^6 \int_0^T \int_\Omega |\nabla v^{1/6}|^6 dt &\leq \liminf_{j \rightarrow \infty} \int_0^T \int_{B_{R_j}} \frac{|\nabla v_j|^6}{v_j^5}. \end{aligned}$$

The weak limits (3.20) and (3.21) give

$$\begin{aligned} \int_0^T \int_\Omega uv |\nabla u|^2 &\leq \liminf_{j \rightarrow \infty} \int_0^T \int_{B_{R_j}} u_j v_j |\nabla u_j|^2, \\ \int_0^T \int_\Omega u^3 v |\nabla v|^2 &\leq \liminf_{j \rightarrow \infty} \int_0^T \int_{B_{R_j}} u_j^3 v_j |\nabla v_j|^2. \end{aligned}$$

The weak L^2 -convergence of the population fluxes and of Δv_j handles their squared norms. For terms depending only on u_j and v_j , first use local almost-everywhere convergence and Fatou's lemma on $\Omega \times (\tau, T)$, and then let $\tau \downarrow 0$. Along a countable exhaustion of $\mathbb{R}^2$, the quartic inequalities hold simultaneously outside one null set of times, so monotone convergence retains the time-essential supremum. Finally let $T \uparrow \infty$. This proves (2.11) and (2.12), including the weak-star step required by the quartic estimate.

Lemma 3.7 gives the $W^{1,\infty}$ -part of (2.2) and the nutrient pointwise bounds. The L^2 -bounds for Δv , v , and uv imply $v \in L^2(0, T; H^2(\mathbb{R}^2))$ and $v_t \in L^2(\mathbb{R}^2 \times (0, T))$.

The local almost-everywhere convergence, the uniform pointwise bound, the ball mass estimate, and a countable exhaustion of $\mathbb{R}^2$ give, for every $T > 0$,

$$\operatorname{ess\,sup}_{0 < t < T} \left(\|u(t)\|_{L^1(\mathbb{R}^2)} + \|u(t)\|_{L^\infty(\mathbb{R}^2)} \right) \leq C.$$

The estimates just proved and the Cauchy–Schwarz inequality give

$$|uv\nabla u - u^2v\nabla v| + uv \in L^1(\mathbb{R}^2 \times (0, \infty)).$$

Lemma 3.8 now supplies an $L^1 \cap L^\infty$ -valued representative of u that is continuous in BL^* . In particular, for $0 \leq t_1 < t_2$,

$$\|u(t_2) - u(t_1)\|_{\text{BL}^*} \leq \int_{t_1}^{t_2} \int_{\mathbb{R}^2} (|uv\nabla u - u^2v\nabla v| + uv). \quad (3.31)$$

This proves the first limit in (2.7). The heat representation gives $v \in C([0, \infty); L^1)$ and the second limit.

Equation (3.28) gives (2.9) for every $t \geq 0$. Integrating (3.23) over $\mathbb{R}^2$ gives (2.8); their sum gives (2.10). This proves the theorem. $\square$

4. POSITIVE-TIME REGULARITY

Proof of Theorem 2.4. The Sobolev assertions for u are part of Definition 2.1 and the construction in Section 3. Fix smooth domains $\Omega_0 \Subset \Omega_1 \Subset \Omega_2 \Subset \mathbb{R}^2$ and $0 < \tau < T$. Theorem 2.2 gives

$$0 < c_0 := \operatorname{ess\,inf}_{\Omega_2 \times [\tau/4, T+1]} v \leq v \leq \|v_0\|_{L^\infty}, \quad 0 \leq u \leq C \quad \text{on } \Omega_2 \times [\tau/4, T+1].$$

Since $uv \in L^\infty$, interior parabolic L^p -estimates for

$$v_t - \Delta v = -uv$$

give

$$v \in W_p^{2,1}(\Omega_1 \times (\tau/2, T+1/2)) \quad \text{for every } 1 < p < \infty.$$

Choose $p > 4$. Parabolic Sobolev embedding yields

$$v \in C^{1+\theta, (1+\theta)/2}(\overline{\Omega_1} \times [\tau/2, T+1/2])$$

for some $\theta \in (0, 1)$.

Define

$$\vartheta(s) = \min \left\{ \max\{s, 0\}, \|u\|_{L^\infty(\mathbb{R}^2 \times (0, \infty))} \right\}.$$

Set

$$\mathbf{a}(x, t, s, \zeta) = |s|v(x, t)\zeta - \vartheta(s)^2v(x, t)\nabla v(x, t), \quad b(x, t, s) = \vartheta(s)v(x, t).$$

On the larger cylinder the population equation agrees with

$$u_t - \nabla \cdot \mathbf{a}(x, t, u, \nabla u) = b(x, t, u).$$

For all $s \in \mathbb{R}$ and $\zeta_1, \zeta_2 \in \mathbb{R}^2$,

$$(\mathbf{a}(x, t, s, \zeta_1) - \mathbf{a}(x, t, s, \zeta_2)) \cdot (\zeta_1 - \zeta_2) \geq c_0 |s| |\zeta_1 - \zeta_2|^2,$$

$$\mathbf{a}(x, t, s, \zeta_1) \cdot \zeta_1 \geq \frac{c_0}{2} |s| |\zeta_1|^2 - C,$$

$$|\mathbf{a}(x, t, s, \zeta_1)| \leq C(|s| |\zeta_1| + |s|^{1/2}),$$

$$|b(x, t, s)| \leq C.$$

In the coercivity estimate, $\vartheta(s)^4/|s|$ is understood to be zero at $s = 0$; elsewhere it is bounded by $\|u\|_{L^\infty}^3$. Here and below, C may depend on the selected cylinders and on the established local bounds for v , ∇v , and u . The maps $\mathbf{a}$ and b are Carathéodory functions. They are locally continuous in (x, t) , uniformly on bounded (s, ζ) -sets, because v and ∇v have the preceding parabolic Hölder regularity.

The weak solution belongs to the required energy class:

$$u \in L^2(\tau/2, T+1/2; H^1(\Omega_1)), \quad u_t \in L^2(\tau/2, T+1/2; H^{-1}(\Omega_1)).$$

The Lions–Magenes lemma gives $u \in C([\tau/2, T+1/2]; L^2(\Omega_1))$, and the boundedness of u further yields

$$u^{3/2}, u^2 \in L^2(\tau/2, T+1/2; H^1(\Omega_1)).$$

Thus the local weak-solution and structural hypotheses of Porzio and Vespi [15, Theorem 1.3 and Remark 1.4] hold with $p = 2$, $m = 2$, and degeneracy function $r \mapsto r$. All lower-order coefficient functions may be chosen constant on the selected finite cylinder, hence belong to $L_t^\infty L_x^\infty$ and satisfy

the stated strict two-dimensional mixed-integrability conditions. The source term has zero gradient coefficient. It follows that

$$u \in C^{\alpha, \alpha/2}(\overline{\Omega_0} \times [\tau, T])$$

for some $\alpha \in (0, 1)$, depending only on the local cylinders and the displayed structural bounds. After replacing α by $\min\{\alpha, \theta\}$, the product uv is locally $C^{\alpha, \alpha/2}$. The interior Schauder estimate for the nutrient equation gives

$$v \in C^{2+\alpha, 1+\alpha/2}(\overline{\Omega_0} \times [\tau, T]),$$

with this possibly smaller exponent. $\square$

5. LARGE-TIME BEHAVIOR

Proof of Theorem 2.5. The heat-kernel estimate from $L^1(\mathbb{R}^2)$ to $L^q(\mathbb{R}^2)$, combined with (3.24), gives (2.13). In particular,

$$\|v(\cdot, t)\|_{L^\infty(\mathbb{R}^2)} \leq Ct^{-1}\|v_0\|_{L^1(\mathbb{R}^2)}, \quad t > 0.$$

The uniform population bound and the preceding L^∞ -decay imply

$$\|v(\cdot, s)\|_\infty + \|u(\cdot, s)v(\cdot, s)\|_\infty \leq \frac{C}{s}, \quad s \geq 1.$$

For $t \geq 2$, insertion into the sliding estimate (3.26) gives $\|\nabla v(\cdot, t)\|_\infty \leq C/(t-1)$. Together with (3.25), this proves (2.14).

Estimate (3.31) shows that $u(t)$ is Cauchy in BL^* as $t \rightarrow \infty$. Every term on the right-hand side of (3.29) tends to zero as $R \rightarrow \infty$, so the population trajectory is tight. The uniform L^∞ -bound also gives

$$\sup_{t>0} \int_E u(x, t) \, dx \leq C|E|$$

for every measurable $E \subset \mathbb{R}^2$. The trajectory is therefore relatively weakly compact in $L^1(\mathbb{R}^2)$.

Let $t_n \rightarrow \infty$. A subsequence converges weakly in L^1 to a nonnegative function u_∞ . The BL^* -Cauchy property shows that all weak cluster points agree on $C_c^\infty(\mathbb{R}^2)$, hence agree almost everywhere. Relative weak compactness and uniqueness of the cluster point therefore give (2.15) for the full trajectory. The uniform L^∞ -bound passes to the weak limit, so $u_\infty \in L^1 \cap L^\infty$. For a sequence $t_n \rightarrow \infty$ along which $u(t_n) \rightharpoonup u_\infty$ in L^1 , the definition of the bounded-Lipschitz dual norm gives

$$\|u(t) - u_\infty\|_{\text{BL}^*} \leq \liminf_{n \rightarrow \infty} \|u(t) - u(t_n)\|_{\text{BL}^*}.$$

Applying (3.31) before taking this lower limit gives

$$\|u(\cdot, t) - u_\infty\|_{\text{BL}^*} \leq \int_t^\infty \int_{\mathbb{R}^2} (|uv\nabla u - u^2v\nabla v| + uv),$$

which proves (2.16). Finally, (2.9) and weak L^1 -convergence give (2.17); (2.8) gives (2.18). $\square$

ACKNOWLEDGMENTS

Ruiyi Zhang and Xiaoxin Zheng were supported by the National Natural Science Foundation of China under Grant No. 12371231.

DATA AVAILABILITY STATEMENT

Data sharing is not applicable to this article because no new data were created or analyzed in this study.

CONFLICT OF INTEREST

The authors declare that they have no conflict of interest.

REFERENCES

- [1] T. Black, S. Kohatsu, and D. Wu, Global solvability and large-time behavior in a doubly degenerate migration model involving saturated signal consumption, *J. Evol. Equ.* **26** (2026), Article 24. doi:10.1007/s00028-025-01163-w.
- [2] V. Calvez and L. Corrias, The parabolic–parabolic Keller–Segel model in $\mathbb{R}^2$, *Commun. Math. Sci.* **6** (2008), 417–447. doi:10.4310/CMS.2008.v6.n2.a8.
- [3] L. Corrias and B. Perthame, Critical space for the parabolic–parabolic Keller–Segel model in $\mathbb{R}^d$, *C. R. Math. Acad. Sci. Paris* **342** (2006), 745–750. doi:10.1016/j.crma.2006.03.008.
- [4] X.-M. De-Ji, A. Huang, and Y. Wang, Stabilization of arbitrary structures in a three-dimensional doubly degenerate nutrient taxis system, *European J. Appl. Math.*, published online (2026), 1–28. doi:10.1017/S0956792526100321.
- [5] F. Herrero-Hervás, Global weak solutions to a doubly degenerate nutrient taxis system on the whole real line, *Nonlinear Analysis* **271** (2026), Article 114137. doi:10.1016/j.na.2026.114137.
- [6] J. F. Leyva, C. Málaga, and R. G. Plaza, The effects of nutrient chemotaxis on bacterial aggregation patterns with non-linear degenerate cross diffusion, *Physica A* **392** (2013), 5644–5662. doi:10.1016/j.physa.2013.07.022.
- [7] G. Li, Large-data global existence in a higher-dimensional doubly degenerate nutrient system, *J. Differential Equations* **329** (2022), 318–347. doi:10.1016/j.jde.2022.05.007.
- [8] G. Li and M. Winkler, Nonnegative solutions to a doubly degenerate nutrient taxis system, *Commun. Pure Appl. Anal.* **21** (2022), 687–704. doi:10.3934/cpaa.2021194.
- [9] G. Li and M. Winkler, Continuous solutions for a two-dimensional cross-diffusion problem involving doubly degenerate diffusion and logistic proliferation, *Anal. Appl. (Singap.)* **23** (2025), 489–510. doi:10.1142/S0219530524500374.
- [10] G. Li and M. Winkler, Proliferation-dominated large time behaviour in a planar doubly degenerate nutrient taxis system, *Proc. Roy. Soc. Edinburgh Sect. A*, published online (2026), 1–30. doi:10.1017/prm.2026.10129.
- [11] T. Matsuyama and M. Matsushita, Fractal morphogenesis by a bacterial cell population, *Crit. Rev. Microbiol.* **19** (1993), 117–135. doi:10.3109/10408419309113526.
- [12] N. Mizoguchi, Global existence for the Cauchy problem of the parabolic–parabolic Keller–Segel system on the plane, *Calc. Var. Partial Differential Equations* **48** (2013), 491–505. doi:10.1007/s00526-012-0558-4.
- [13] M. Ohgiwari, M. Matsushita, and T. Matsuyama, Morphological changes in growth phenomena of bacterial colony patterns, *J. Phys. Soc. Japan* **61** (1992), 816–822. doi:10.1143/JPSJ.61.816.
- [14] X. Pan, Superlinear degradation in a doubly degenerate nutrient taxis system, *Nonlinear Anal. Real World Appl.* **77** (2024), Article 104040. doi:10.1016/j.nonrwa.2023.104040.
- [15] M. M. Porzio and V. Vespi, Hölder estimates for local solutions of some doubly nonlinear degenerate parabolic equations, *J. Differential Equations* **103** (1993), 146–178. doi:10.1006/jdeq.1993.1045.
- [16] R. G. Plaza, Derivation of a bacterial nutrient-taxis system with doubly degenerate cross-diffusion as the parabolic limit of a velocity-jump process, *J. Math. Biol.* **78** (2019), 1681–1711. doi:10.1007/s00285-018-1323-x.
- [17] B.-N. Tran and J. Yang, Global weak solvability for doubly degenerate nutrient taxis system in physical dimension, arXiv:2508.03268, 2025, revised 2026. arXiv:2508.03268.
- [18] M. Winkler, Does spatial homogeneity ultimately prevail in nutrient taxis systems? A paradigm for structure support by rapid diffusion decay in an autonomous parabolic flow, *Trans. Amer. Math. Soc.* **374** (2021), 219–268. doi:10.1090/tran/8163.
- [19] M. Winkler, Small-signal solutions of a two-dimensional doubly degenerate taxis system modeling bacterial motion in nutrient-poor environments, *Nonlinear Anal. Real World Appl.* **63** (2022), Article 103407. doi:10.1016/j.nonrwa.2021.103407.
- [20] M. Winkler, Stabilization of arbitrary structures in a doubly degenerate reaction–diffusion system modeling bacterial motion on a nutrient-poor agar, *Calc. Var. Partial Differential Equations* **61** (2022), Article 108. doi:10.1007/s00526-021-02168-2.
- [21] M. Winkler, L^∞ bounds in a two-dimensional doubly degenerate nutrient taxis system with general cross-diffusive flux, *J. Differential Equations* **400** (2024), 423–456. doi:10.1016/j.jde.2024.04.028.
- [22] D. Wu, Refined existence theorems for doubly degenerate chemotaxis–consumption systems with large initial data, *NoDEA Nonlinear Differential Equations Appl.* **31** (2024), Article 104. doi:10.1007/s00030-024-00991-y.
- [23] Y. Xiang, Boundedness and large-time behavior in a two-species doubly degenerate diffusion chemotaxis system with logistic proliferation, *Nonlinear Anal. Real World Appl.* **89** (2026), Article 104525. doi:10.1016/j.nonrwa.2025.104525.
- [24] Z. Zhang and Y. Li, Global boundedness for a two-dimensional doubly degenerate nutrient taxis system, arXiv:2411.01133, 2024. arXiv:2411.01133.
- [25] Z. Zhang and Y. Li, Boundedness in a two-dimensional doubly degenerate nutrient taxis system, *Math. Models Methods Appl. Sci.* **36** (2026), 527–559. doi:10.1142/S0218202526500077.

SCHOOL OF MATHEMATICAL SCIENCES, BEIHANG UNIVERSITY, BEIJING 100191, P. R. CHINA
 Email address: ruiyizhang@buaa.edu.cn

SCHOOL OF MATHEMATICAL SCIENCES, BEIHANG UNIVERSITY, BEIJING 100191, P. R. CHINA
 Email address: xiaoxinzheng@buaa.edu.cn